THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · HARD LEVEL CHAPTER TEST SET 2

Chapter 4: **Carbon and its Compounds** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. On complete combustion in air, ethanol gives: 1

(a) ethanoic acid only **(b)** ethene and water

(c) carbon dioxide and water **(d)** carbon monoxide and hydrogen

2. A compound with the formula $C_3H_6O_2$ smells fruity and hydrolyses in alkali to give sodium ethanoate. It is: 1

(a) propanol **(b)** propanoic acid **(c)** methyl ethanoate **(d)** ethyl methanoate

3. Assertion (A): Ethanoic acid is called a weak acid. Reason (R): It ionises only partially in water. 1

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

4. The percentage of carbon in methane (C = 12, H = 1) is: 1

(a) 80% **(b)** 60% **(c)** 75% **(d)** 25%

5. A student burns two compounds P and Q. P gives a blue clean flame and Q gives a yellow sooty flame. Q is probably: 1

(a) unsaturated **(b)** water **(c)** saturated **(d)** an alcohol

6. Assertion (A): Carbon forms a large number of compounds. Reason (R): Carbon exhibits catenation and tetravalency. 1

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

7. The percentage of carbon in ethene is: 1

(a) 92.3% **(b)** 85.7% **(c)** 75% **(d)** 80%

8. In a micelle, the central core is composed of: 1

(a) calcium ions **(b)** water

(c) carboxylate heads **(d)** hydrocarbon tails

9. **Assertion (A):** Vegetable oils are hydrogenated using nickel catalyst to obtain saturated fats. 1
Reason (R): Hydrogenation converts the single bonds of unsaturated oils into double bonds.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

10. **Assertion (A):** Diamond is a good conductor of electricity. 1
Reason (R): In diamond each carbon atom uses all four of its valence electrons to form bonds with four other carbon atoms.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Name the compound of formula $C_4H_8O_2$ formed from ethanol and ethanoic acid, and its smell. 2

12. Explain why methane burns with a non-sooty flame while ethyne burns with a sooty flame. 2

13. Why does a sooty flame indicate incomplete combustion? Which has the highest soot: ethane, ethene or ethyne? 2

14. Why is ethanol not used directly as a fuel in vehicles in pure form in many places? (consider its properties) Give one advantage. 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. Write the equations in the conversion: ethene -> ethane -> chloroethane and name the reaction types. 3

16. An ester has formula $C_4H_8O_2$ and on hydrolysis gives ethanoic acid. Identify the ester and the alcohol. 3

17. Explain how ethanol is prepared from ethene indirectly? Or give industrial method (hydration) with equation. 3

OR

An organic compound X ($C_2H_4O_2$) on reaction with ethanol in conc. H_2SO_4 gives a sweet-smelling product. Identify X, the product and give the equation. 3

18. How will you distinguish between 3
(a) ethane and ethene
(b) ethanol and ethanoic acid?

OR

State two reasons why detergents are preferred to soaps for washing clothes in hard water. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** In the school laboratory, Priya added a pinch of sodium hydrogencarbonate to a test tube containing ethanoic acid (vinegar-type acid) and observed brisk effervescence. In another test tube, her teacher warmed ethanoic acid with ethanol and a few drops of concentrated sulphuric acid and then poured the mixture into water; a sweet fruity smell was noticed. 5
(i) Name the gas evolved in the first test tube.
(ii) How can the gas be tested?
(iii) Name the organic product formed in the second test tube.
(iv) Write the chemical equation for the second reaction and name the reaction.

20. Explain the cleansing action of soaps in terms of micelle structure; state why a detergent is better in hard water; give the structure of a typical soap and detergent. 5

OR

Describe the reaction of ethanoic acid with sodium carbonate, with the equation, observation and the test for the gas. 5

END OF QUESTION PAPER Total: 40 Marks